

Preliminary Technical Specification

Let'sDataTalk — club operations platform for grassroots football (AU/NZ)

Pilot club: North Star FC · Vendor: Let'sDataTalk · v1 preliminary — **internal use, not anonymised**

Internal document. Contains real club, federation, and third-party platform names, and correspondence with Football Queensland. Do not circulate outside the team without generating the anonymised version.

Preliminary document built from the project's enterprise architecture repository (strategy and business layers, 13 scope documents, 5 decision records, and a running log of 45 open questions), from two rounds of client answers (July and August 2026), and from the email exchange with Football Queensland's Registrations team dated 31 July and early August 2026. It is a basis for reading effort. **No code exists yet:** the project is pre-MVP and everything described here is validated architecture, not implementation.

Notes before you start

- **The main blocker is legal, not technical, and it is recent.** Football Queensland stated in writing (31 July 2026) that Squadi's terms of use *do not permit club accounts or Squadi data to be connected or integrated with unauthorised third-party systems*. That does not merely close the API: it reaches the **CSV fallback** the entire reconciliation capability (C14) was designed on — which is also the product's commercial wedge. Until it resolves, C14 does not get built. It is the most expensive open item in the project.
- **Confirmed in writing — there is no club-level API, and there will not be one by that route.** Access is restricted to Football Australia, Football Queensland, and «approved system partners». The path to stage 2 is not a technical request but **approved partner status**: a commercial and contractual move by the vendor, not something the club can ask for on its behalf.
- **Confirmed with the client (negotiations that delete work):** Squadi is the authoritative system of record and the platform never writes back to it; calendar sync is **one-way**, with no OAuth and no writing into personal accounts; carnival and event management is **deferred** past the MVP; no state voucher programme exposes an API usable at the pilot club's scale. Each of these lines removes weeks of work.
- **Not confirmed — do not treat as settled:** there is no client statement about **offline operation** in the field (match day, grounds without signal). This is not «not applicable»; it is a gap. Target devices and expected service levels are likewise undefined.
- **Scale:** the pilot club handles **~800 players and ~700 registrations per season**. Multi-tenant from day 1 by principle (P5), not as a later optimisation. The decision to **replace the incumbent** (Majestri) turns every sale into a mid-season data migration.
- **Compliance:** Privacy Act 1988 and the Australian Privacy Principles, New Zealand's Privacy Act 2020, state working-with-children screening regimes (Blue Card in Queensland), and FIFA/Football Australia's international transfer process. GDPR was adopted as a **voluntary design standard**, not a binding obligation — see section 10.

1. Context and problem

Grassroots football clubs in Australia and New Zealand run player and referee administration spread across the incumbent system Majestri, the federation's mandated platforms (Squadi and PlayFootball), spreadsheets, forms, email, WhatsApp, and Drive. The cost is measurable: registering a player **takes weeks**, and the dominant cause confirmed by the client is guardian confusion at capture — typically the child's nickname instead of their legal name, which the operator processes as a mismatch and the registration re-loops. As at 15 May 2026 the pilot club had **799 players, 707 registrations, and 49 incomplete**, with **78 missing from PlayFootball and 40 missing from Squadi**. Under Football Queensland policy those 40 cannot take the field: the latency is not administrative, it costs playing time.

2. System objective

Unify a club's registration, finance, documents, referee management, competitions, and community events around a **single Person identity**, so that one individual participates as player, referee, coach, guardian, or committee member without duplicated records. Commercially, replace the incumbent system rather than complement it.

3. Scope

- **Identity and roles** — one Person, many roles with effective dates; duplicate detection with human confirmation.
- **Player registration** — season, guardians, required documents, registration states, and the international transfer (ITC) path for players previously registered with another federation.
- **Player finance** — fees, payment plans, instalments, state government vouchers subject to prior committee approval, reconciliation.
- **Full referee lifecycle** — classification and pathway, requirements, availability, appointments, role-conflict detection, post-match verification, and payment.
- **Data quality and compliance** — deterministic rules engine, exception lists, audit trail.
- **Communications and dashboards** — templated transactional messaging; registration, finance, and refereeing dashboards.

- **Historical consolidation** — read-only extraction of the pilot club's ≥ 3 years into a unified model.
- **Multi-tenant operations** — provisioning, role-based access control, per-club configuration and branding.
- **Competitions and calendar, calendar distribution** by iCalendar subscription, and **consent and privacy rights**.
- **External registration reconciliation (C14)** — blocked see section 11, risk 1.

Explicitly out of scope: writing into federation systems (forbidden by the platform's own principle until a formal exception exists); two-way calendar sync; carnival and event management in the MVP; replacing Squadi as the system of record — *not possible*, the federation mandates it and player eligibility depends on it.

4. Users and roles

Role	Who they are	What they can do	What they cannot do
Registrar	Club administrator, usually a volunteer	Capture and complete registrations; approve registration exceptions; resolve duplicate identity candidates	See another club's data; apply vouchers; change competition rules
Treasurer / Finance Admin	Club finance officer	Issue and adjust invoices and payment plans; apply approved vouchers; execute referee payment batches	Approve the voucher programme (that is the committee's); approve their own payment
Committee	The club's governing body	Approve joining a state voucher programme; approve club policy	Touch invoices; execute payments
Referee coordinator	Appoints officials to matches	Propose and confirm appointments; see conflicts and warnings; receive vacancy notifications	Appoint past an eligibility block; alter a referee's classification
Referee / match official	Includes minors from entry-level classifications	Declare availability; accept or decline with a reason; subscribe to their calendar; claim payment for a verified match	See others' appointments; change the authoritative record from a personal calendar
Guardian	Parent or responsible adult for a minor	Register the minor; grant and revoke consents; receive a minor official's calendar	See other families' data; consent on behalf of a player aged 18 or over
Assistant (AI)	Software actor, not human	Draft, summarise, classify, flag for human review	Never approve documents, reject players, change debts, approve payments, or promote referees

Two implications that do not fit in cells. **Isolation between clubs is a principle, not a setting:** no club's data is visible to another, with a single scoped and documented exception — the published schedule and results of a multi-club community event, visible without an account. And **the Assistant's «cannot» column is a recorded decision**, not a temporary technical limitation: the AI's autonomy level is fixed by architecture decision, and changing it is a change of scope.

5. User stories (draft, not validated story by story with the client)

#	Story	Acceptance criteria	Priority
1	As a guardian, I want to enter my child's details once and have the club validate them, so I do not have to re-do the registration in Squadi	The form requires the legal name and captures the everyday name separately; rejects incomplete fields before submission; the club receives the registration marked ready, or with the gap named	Non-negotiable
2	As a registrar, I want to see which of our players are absent from Squadi, so I can resolve them before the season starts	Dated list carrying the age of its source; each absence is presented as an eligibility exception, not as a data-difference row	High — blocked
3	As a coordinator, I want a referee whose screening check has expired removed from future appointments automatically	Withdrawal reaches every future commitment; the holder and the coordinator are notified with the vacancy to fill; past records are untouched	High
4	As a referee, I want my appointments in the calendar I already use, so I am not dependent on WhatsApp messages	Subscription by a private, revocable URL; contains only that person's own commitments; no third parties' personal data	Medium
5	As a guardian, I want to decide separately whether I allow my child's image to be used for promotion	Consent independent of the registration, off by default, granular per channel, and revocable; refusing it does not block or delay the registration	High
6	As a treasurer, I want to apply an approved state voucher to a family's invoice and claim the reimbursement	Only committee-approved programmes; code verified beforehand; impossible to apply twice or claim twice	Medium

6. Data model (main entities, draft)

- **Person** — the single identity. Carries **two first-class names**: the legal one, as it appears on the passport or birth certificate, which is what external registration and cross-system matching use; and the everyday one, which is what any human sees. May carry an identification photograph. Separating the two is not cosmetic — it is the direct correction to the dominant cause of registration delay.
- **Person Role** — a role scoped to a club and season, with effective dates. This is what lets one person be a player and a referee without duplicating.
- **Player Registration** — season enrolment with a status. The «pending external registration» state is an **eligibility gate**, not paperwork: without it there is no playing time.
- **Consent** — **three distinct types**, not one: a collection notice for the registration, an identification-photograph consent, and a publicity consent. Each with its own purpose, scope, date, and revocation. Modelling this as a boolean is the expensive mistake in this entity.
- **Working with Children Check** — validity and status. Depends on the club being *linked* as the responsible organisation in the state register, which is what enables change notifications.
- **Match Official Appointment** and **Appointment Response** — the response requires a reason on decline or withdrawal.
- **External System Extract** — a CSV report from Squadi or PlayFootball carrying *its own declared limitations* and its **authorisation basis**. If the basis is unconfirmed, it is not ingested. Blocking
- **Competition, Playing Format, and Competition Regulation catalogues** — **configuration data, not code**. They vary by state federation and by season; fixing them in code is the decision that costs a rewrite later.
- **Voucher Programme** — six state schemes with different values, frequencies, and eligibility; also configuration, not code.

7. Main flows

- **Registration:** guardian enters once → deterministic validation (legal name, guardian, documents, payment) → registration marked «Squadi-ready» → entry into Squadi → player eligible.
- **Reconciliation:** external system extract → matching with no shared identifier (name + date of birth + email) → candidates for human confirmation → dated eligibility exceptions. Blocked
- **Referee appointment:** availability → proposal → conflict rules (blocks and audited warnings) → acceptance or decline with reason → calendar → match → verification → claim → payment batch.
- **Screening-check lifecycle:** onboarding and linkage in the state register → register notifies on status change → automatic withdrawal from every future commitment → vacancy notice to the coordinator. Annual reconciliation pass at the start of the season.
- **Consent and erasure:** capture per purpose → revocation → erasure request assessed against lawful basis → deletion or de-identification, with the reason for any refusal recorded.

8. Required integrations

System	Direction	What is exchanged / status
Squadi (Football Queensland)	External → platform, by CSV	Registration and user reports for reconciliation. Blocked pending clarification of the terms of use
Squadi	Platform → external	Assisted submission of prepared registrations. Not available: requires approved partner status and an exception to the platform's own no-write principle
PlayFootball / Football Australia	External → platform, by CSV	Equivalent to the above; formats not yet documented
Majestri	External → platform, one-off	Club data migration on replacement. Bulk export feasibility unconfirmed
Square	Two-way	Collections and payment reconciliation. Adopted first provider
State screening register	External → platform	Notification of a credential's status change, via organisation linkage
Calendar clients	Client pulls from the platform	Per-person iCalendar feed, private and revocable; no OAuth and no third-party credentials

Not integrated: ERP, payroll, accounting (Xero remains a post-MVP priority), WhatsApp, Strava, or any state voucher programme API — none exposes access usable at this scale. Voucher claiming is portal- or CSV-based with human intervention. Changing any of these lines is a change of scope.

9. Non-functional requirements

Requirement	Definition
Multi-tenancy	Strict per-club isolation from day 1, with one scoped and documented exception for published multi-club event content. An architecture principle, not a setting
Offline operation	Not specified by the client. Do not assume it does not apply: match-day capture happens at grounds that may have no signal. Must be closed before the mobile architecture is fixed
Target devices	Not specified. Real use is guardians and volunteers on their own phones, mixed range. Implies QA on real hardware, not just an emulator
Security	Role-based access control. Calendar feed URLs are bearer credentials: they must be unguessable and immediately revocable
Audit	Every override of a conflict warning is audited. The reason for refusing an erasure request is recorded
Retention	Three years as stated by the client, with a caveat: Australian statutory minimums may exceed it for financial and child-safety records
Configurability	Fee schedules, thresholds, playing formats, competition regulations, voucher values, and carnival conditions are data, not code
Backup and service level	Not specified by the client — standard proposal pending
Performance	Not specified. The real peak is predictable and sharp: registration opening before the season starts

10. Regulatory requirements

The Privacy Act 1988 and the Australian Privacy Principles apply in Australia, the Privacy Act 2020 in New Zealand, state working-with-children screening regimes, and FIFA/Football Australia regulations on the international transfer of minors.

What can already be designed: explicit, purpose-scoped, revocable consent before a minor's data is processed; three separate consents (registration, identification photograph, publicity) where publicity is optional and off by default; an erasure protocol honoured unless a named lawful basis applies, with de-identification where deletion is unlawful and **the basis recorded on every refusal**; continuous screening verification by state-register notification, with automatic withdrawal from future commitments on expiry or revocation; validity tested against *the end of the season* rather than today's date; and a privacy framework configured **per tenant**, not globally.

Pending confirmation, do not treat as closed: whether the three-year period is lawful for every record class — a deletion job built on the current assumption could destroy records that must be kept; at what age a young person assumes control of their own consent (18 is fixed for publicity only, and a 16- or 17-year-old referee currently has none); whether state registers notify on the terms assumed, and whether the club is in fact linked for every child-related role; and legal review of the consent wording, which is

Framing note: **GDPR does not bind this operation** while it processes no EU residents' data. The APPs contain no general right to erasure — they require destruction or de-identification once data is no longer needed, and give a right of *correction*. The adopted erasure protocol **exceeds** what AU/NZ law requires: a defensible design decision, not an obligation met, and it is recorded that way.

11. Technical risks

#	Risk	Impact	Mitigation
1	Squadi's terms of use may prohibit ingesting the CSV the club exports from its own account	Removes C14, which is both the most valuable capability and the commercial argument for displacing the incumbent. The product would be <i>worse</i> than the system it aims to replace, on the axis chosen to beat it	Already asked of Football Queensland in the open thread. Standing rule: an unconfirmed authorisation basis counts as not permitted — the ingestion is not built. Cheap parallel test: Majestri already performs that same comparison, in the open; find out under what arrangement
2	The club stated in writing to the federation that its system «has no connection to Squadi and holds no Squadi data»	True today and for stage 1; false for C14 as designed. Building it before the reply would make a written statement to the governing body untrue	Treat the statement as an active constraint, not a courtesy. Any ingestion design waits for the reply
3	Three rules in tension: three-year retention, statutory minimums, and the right to erasure	Both a deletion job and an erasure workflow could destroy records that must legally be kept	One legal answer settles all three. Build neither flow before it exists
4	«Automated» screening verification may not exist as assumed	If the state register does not notify, continuous verification degrades to «whenever someone remembers» and automatic withdrawal fires late — precisely the scenario the rule exists to prevent	Confirm the linkage and notification mechanism with the state register <i>before</i> designing the process. Also check how many current holders are unlinked: if they were never linked, no notification arrives today and nobody knows it
5	Identity matching with no shared key between systems	A false positive attaches one child's registration, payments, and eligibility to a different child	Every match is a candidate for human confirmation, never an automatic merge. Mandatory legal names improve the signal directly
6	Migration from the incumbent without confirmed bulk export	Every sale is a mid-season migration; if export is restricted, switching cost rises sharply and the replacement strategy loses its practical route	Confirm export feasibility before committing to a commercial timeline
7	Competition and format catalogues built as fixed values	They vary by state federation and season; fixing them forces a rewrite at the second client	Configuration from day one, already declared in the data model
8	Image volumetry estimated without the external specification	Recurring storage cost miscalculated and, worse, photos rejected by the federation on format — reintroducing the guardian re-loop	Request the exact specification in the same open thread. Meanwhile, capture to the strictest standard

12. Estimation by block (relative complexity, no prices or hours)

Functional block	Complexity	Note
Base platform and multi-tenancy	High	Strict isolation plus a scoped public exception, role-based control, and per-club configuration. Paid once, and it conditions everything else
Identity and roles	Medium-High	One person with overlapping, dated roles, plus duplicate detection with human confirmation and the legal/everyday name pair
Player registration	Medium-High	Deterministic validation and states; the eligibility gate and the ITC path add rules, not just forms
Player finance and vouchers	Medium-High	Payment plans plus six configurable state schemes with committee approval, verification, and reimbursement
Referee lifecycle and payment	High	The block with the most business rules in the system: blocking conflicts versus audited warnings, classification pathway, post-match verification, and the payment chain
Consent and privacy	Medium	Three independent consents, revocation, erasure with recorded lawful basis. Medium rather than low because it cuts across every entity

Screening verification	Medium	Needs an <i>inbound</i> path for register notifications — an event received, not a scheduled job. A different integration shape from the rest
Calendar by subscription	Low	One standard artifact serves every calendar client; no OAuth, no per-vendor integration. The best value-to-effort block in the project
Communications and dashboards	Medium	Transactional templates and per-area dashboards; real scope depends on what counts as parity with the incumbent
Historical consolidation	Medium	Read-only extraction of the pilot club's three years; its value is de-risking everything else by profiling real data early
External reconciliation (C14)	High	Blocked. Matching without a shared key, extracts with declared blind spots, and results that go stale. Do not schedule until risk 1 resolves
Competitions, calendar, and events	Medium	The competitions part is reference data; community events are deferred past the MVP

13. Volumetry and hard data

What was measured and where it came from. From the pilot club's comparison screen (2026 season, as at 15 May 2026): **799 players, 707 registrations, 49 incomplete**. Against PlayFootball: 721 matched, **78 missing**. Against Squadi: 759 matched, **40 missing**. Both comparisons were last run on 15 May and were still being read some **ten weeks later** — the figure is real and it also illustrates the problem: a gap list is a snapshot, and undated it invites decisions on facts that have already moved.

Storage projection per club per season. The federation's image specification is *unconfirmed*, so the average size is an assumption and it is the variable that moves the result most:

- Identification photographs: **800 × 200 KB = 156 MB**. Low bound at 120 KB: 94 MB. High bound at 300 KB: 234 MB.
- Identity documents (assumption: **25%** of players require a scanned birth certificate or passport, ~500 KB): **200 × 500 KB = 98 MB**.
- Screening-check evidence (assumption: **120** coaches and volunteers, ~300 KB): **35 MB**.
- **Total per club per season: ~290 MB** in the middle scenario.

Aggregated at scale. At 90 clubs: **~25 GB per season**. With the stated three-year retention, steady state sits around **~76 GB** of binary material, plus the database. That is a manageable figure and it is not what drives infrastructure cost — the point of bringing it is different: **two unconfirmed assumptions move it by a factor of three** (the image specification and the share of players uploading a document), and the retention period may not be lawful as stated. All three close by asking, not by building.

14. Intellectual property and ownership

Element	Whose	Limits to resolve
Reusable base architecture and multi-tenant platform	Let'sDataTalk	This is the asset sold to other clubs; it should be explicit that the pilot club acquires no rights over it by virtue of being first
Club-specific configuration and content	North Star FC	Club rules, templates, catalogues, and branding
Captured data (people, registrations, finance)	North Star FC, as the entity responsible for it	Portability on exit and a deletion commitment at end of service
The ≥3 years of historical data provided for validation	North Star FC	Unresolved: it is used to validate a product subsequently sold to third parties. Permitted use and any required de-identification should be in writing

What to close before signing. There is an overlap of roles worth documenting rather than leaving implicit: the same person participates as an officer of the pilot club and as part of the vendor. That is not a problem in itself, and it explains why discovery was so rich, but it leaves three things undefined: who decides when the club's interest and the product's interest diverge; on what terms the pilot club receives the service (free, preferential rate, temporary exclusivity); and whether any restriction applies to selling to rival clubs in the same federation. These are easy to agree verbally and hard to draft afterwards.

Preliminary internal document. Contains real names and third-party correspondence; it is not a commercial offer and not legal advice. Complexity shown is relative to the problem, not a price.